

Pythagoras' Theorem & Trigonometry

Answer sheet · 27 marks total

Section A — Pythagoras' theorem

A1. $c^2 = 5^2 + 12^2 = 25 + 144 = 169$ (1) $c = \sqrt{169} = \mathbf{13\text{ cm}}$ (1)

A2. $a^2 = 15^2 - 9^2$ (1) $= 225 - 81 = 144$ (1) $a = \sqrt{144} = \mathbf{12\text{ cm}}$ (1)

A3. $\text{diagonal}^2 = 7^2 + 24^2 = 49 + 576 = 625$ (1) $\text{diagonal} = \sqrt{625} = \mathbf{25\text{ cm}}$ (1)

Section B — SOH CAH TOA basics

B1. The **longest side**, opposite the right angle (1)

B2. **SOH**: $\text{opposite} = \text{hypotenuse} \times \sin \theta$ (1)

B3. **TOA**: $\theta = \tan^{-1}(\text{opposite} \div \text{adjacent})$ (1)

Section C — Finding missing sides

C1. $\sin 42^\circ = x \div 10$ (1) $x = 10 \times \sin 42^\circ$ (1) $= \mathbf{6.7\text{ cm}}$ (1)

C2. $\cos 55^\circ = 8 \div h \rightarrow h = 8 \div \cos 55^\circ$ (1) $= \mathbf{13.9\text{ cm}}$ (1)

C3. $\tan 33^\circ = x \div 6 \rightarrow x = 6 \times \tan 33^\circ$ (1) $= \mathbf{3.9\text{ cm}}$ (1)

Section D — Finding missing angles

D1. $\sin \theta = 6 \div 13 = 0.4615 \rightarrow \theta = \sin^{-1}(0.4615)$ (1) $= \mathbf{27.5^\circ}$ (1)

D2. $\tan \theta = 9 \div 5 = 1.8$ (1) $\theta = \tan^{-1}(1.8)$ (1) $= \mathbf{60.9^\circ}$ (1)

Section E — Applying Pythagoras & trigonometry

E1. Use Pythagoras: $\text{height}^2 = 4.5^2 - 1.8^2 = 20.25 - 3.24 = 17.01$ (1) $\sqrt{17.01} = \mathbf{4.1\text{ m}}$ (1)

E2. $\tan 58^\circ = h \div 6$ (1) $h = 6 \times \tan 58^\circ$ (1) $= \mathbf{9.6\text{ m}}$ (1)

Marking note: award method marks for a correctly set-up equation or calculation even if the final answer contains a rounding or arithmetic slip. Accept any correct rounding to 1 decimal place unless a different level of accuracy is specified. Missing units on an otherwise-correct numerical answer should lose the final accuracy mark, not the method mark.